

4. **The Art of Software Testing** – Glenford J. Myers, Corey Sandler, and Tom Badgett, **3rd Edition**, Wiley, 2011.
5. **Effective Software Testing: A Developer's Guide** – Maurício Aniche, Manning Publications, 2022.

Programme : B.Sc. – Information Technology (2026 - 27)			Semester V
Course :Artificial Intelligence			Code : UIARI305
Teaching Scheme		Evaluation Scheme	
Lecture (Hours per week)	Credit	Continuous Assessment (CA) (Marks - 40)	Semester End Examinations (SEE) (Marks- 60 in Question Paper)
3	3	40 Marks	60 Marks
Learning Objectives			
1. Understand the fundamentals of Artificial Intelligence (AI) 2. Learn about intelligent agents 3. Analyze problem-solving techniques in AI 4. Explore knowledge representation and reasoning. 5. Introduce neural networks and their applications. 6. Familiarize with AI applications.			
Learning Outcomes:			
1. The learner will gain knowledge of Artificial Intelligence and its foundations 2. The learner will identify basic principles of AI in problem solving 3. The learner will apply problem-solving methods and search strategies to AI problems 4. The learner will represent and infer knowledge using logical frameworks 5. Learner will be able to design and evaluate AI-based solutions. 6. Learner will be able to demonstrate the ability to apply AI techniques to real-world problems			
Module	Module Content		Duration
I	Artificial Intelligence (AI), AI Perspectives: Acting and Thinking humanly, Acting and Thinking rationally History of AI, Applications of AI, The present state of AI, Ethics in AI. Introduction of agents , Structure of Intelligent Agent, Characteristics of Intelligent Agents. Types of Agents: Simple Reflex, Model Based, Goal Based, Utility Based Agents.		15

	Environment Types: Deterministic, Stochastic, Static, Dynamic, Observable, Semi-observable, Single Agent, Multi Agent Solving Problems by Searching Uninformed Search: Depth First Search, Breadth First Search, Informed Search: Heuristic Function, Greedy Best First Search, A* Search, Local Search: Hill Climbing Search Optimization: Genetic Algorithm, Mini-max Search	
II	Introduction to Evolutionary Computation: Elements of Genetic Algorithms, genetic programming concepts, Evolutionary programming, swarm intelligence Genetic Algorithms and Traditional Search Methods, Some Applications of Genetic Algorithms . Knowledge and Reasoning Definition and importance of Knowledge, Issues in Knowledge Representation, Knowledge Representation Systems, Properties of Knowledge Representation Systems	15
III	Propositional Logic (PL): Syntax, Semantics, Formal logic-connectives, truth tables, tautology, validity, well-formed-formula, Introduction to logic programming (PROLOG) Predicate Logic: FOPL, Syntax, Semantics, Quantification, Inference rules in FOPL, Forward Chaining, Backward Chaining and Resolution in FOPL. Introduction to ANN Units in neural networks, Network structures, Single layer feed-forward neural networks (perceptrons) Applications of ANN Pattern Classification, Associative memories, Optimization, Vector quantization, Control applications, Application in Speech and Image Processing	15
	Total Lectures	45

Books and References:

Sr. No .	Title	Author/s	Publisher	Edition	Year
1.	Artificial Intelligence- A Modern Approach	Stuart Russel, Peter Norvig	Pearson Education	4 th	2020
2.	Artificial Intelligence	Saroj Kaushik	Cengage	1 st	2019

Useful Links

- 1 An Introduction to Artificial Intelligence - Course (nptel.ac.in)
- 2 NPTEL
- 3 <https://www.classcentral.com/course/independent-elements-of-ai-12469>
- 4 <https://tinyurl.com/ai-for-everyone>

Program: B.Sc. – Information Technology (2026 - 27)		Semester: V	
Course: Artificial Intelligence LAB		Course Code: UIARI305P	
Teaching Scheme		Evaluation Scheme	
Lecture (Hours per week)	Credit	Continuous Assessment (CA) (Marks - 20)	Semester End Examinations (SEE) (Marks- 30 in Question Paper)
2	1	20	30
Learning Objectives:			
☐ To understand the basic concepts and techniques used in Artificial Intelligence problem solving. ☐ To learn different search strategies including uninformed and informed search methods. ☐ To understand game playing strategies using algorithms such as the MiniMax algorithm. ☐ To explore optimization and evolutionary techniques such as Genetic Algorithms. ☐ To gain practical knowledge of machine learning concepts by implementing Neural Networks and simple AI chatbots.			
Course Outcomes:			
☐ Students will be able to implement uninformed and informed search algorithms to solve AI problems. ☐ Students will be able to apply the MiniMax algorithm for game simulations such as Tic-Tac-Toe. ☐ Students will be able to implement optimization techniques like Genetic Algorithms for problem solving. ☐ Students will be able to develop simple Neural Network models using Python with and without built-in libraries. ☐ Students will be able to design basic AI-based conversational agents using AIML packages.			

Outline of Syllabus:		
	List of Practicals :	Duration (Lecture)
	1. Implementation of any uninformed search methods. 2. Simulation of TIC-TAC-TOE using MiniMax algorithm 3. Implementation of informed search methods. 4. Implementation of genetic algorithms methods. 5. Implementation of a simple Neural Network using python's inbuilt libraries. 6. Implementation of a simple Neural Network without using python inbuilt libraries. 7. Creating a simple bot using AIML package.	
Total Lectures		30

Program: B.Sc. – Information Technology (2026 - 27)	Semester: V
Course: Security in Computing	Course Code: UISIC306